Phase 2

Task 2 Write Up:

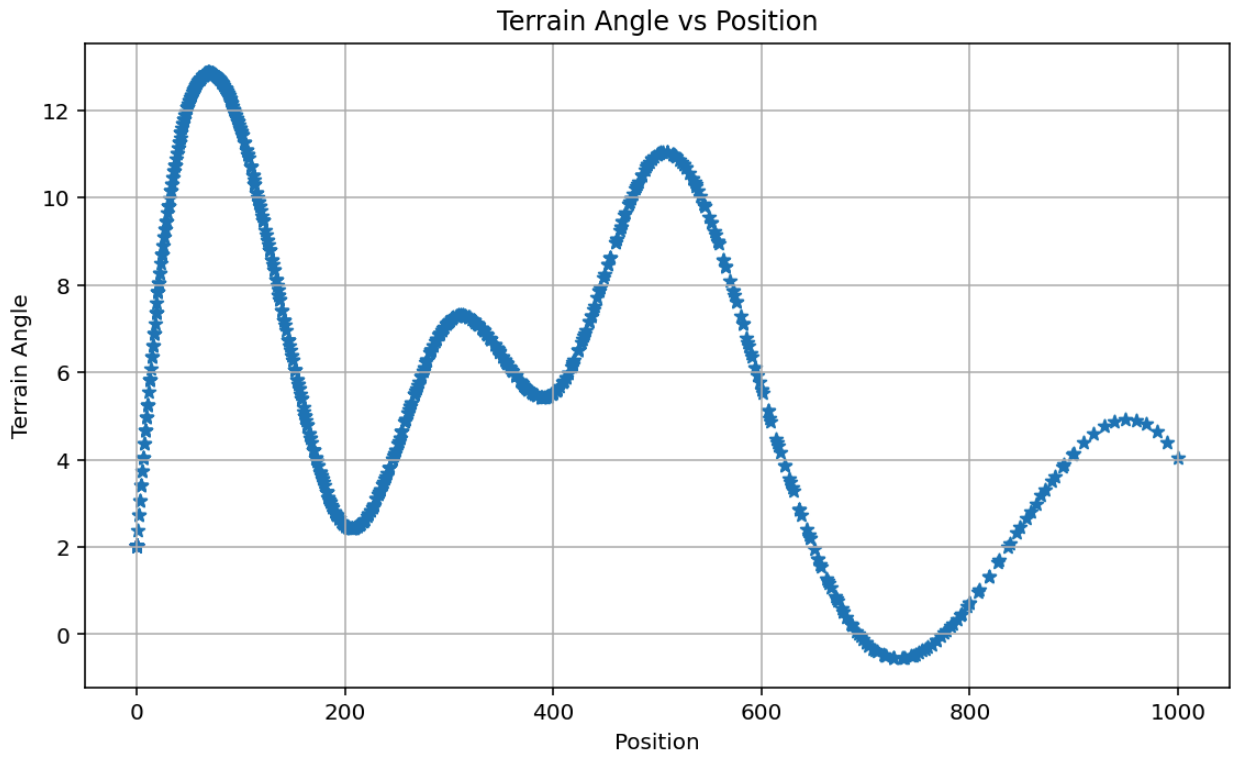


Figure 1: Terrain Angle vs Position Plot

- The curve is relatively smooth, as defined by define_experiment the distance was evaluated from 0.0 to 1000 with 100 as the step size giving the interpolation function enough points to make a smooth scatter plot. The interpolation's function uses existing values and approximates the values in between while preserving the initial data concluding that the curve goes through all known data points. The team observed that some intervals on the graph had more data points concentrated and concluded that alpha_deg was the cause due to the array having its existing data points having wildly different spacing in between each value not giving a completely smooth graph.

Task 5 Write Up:

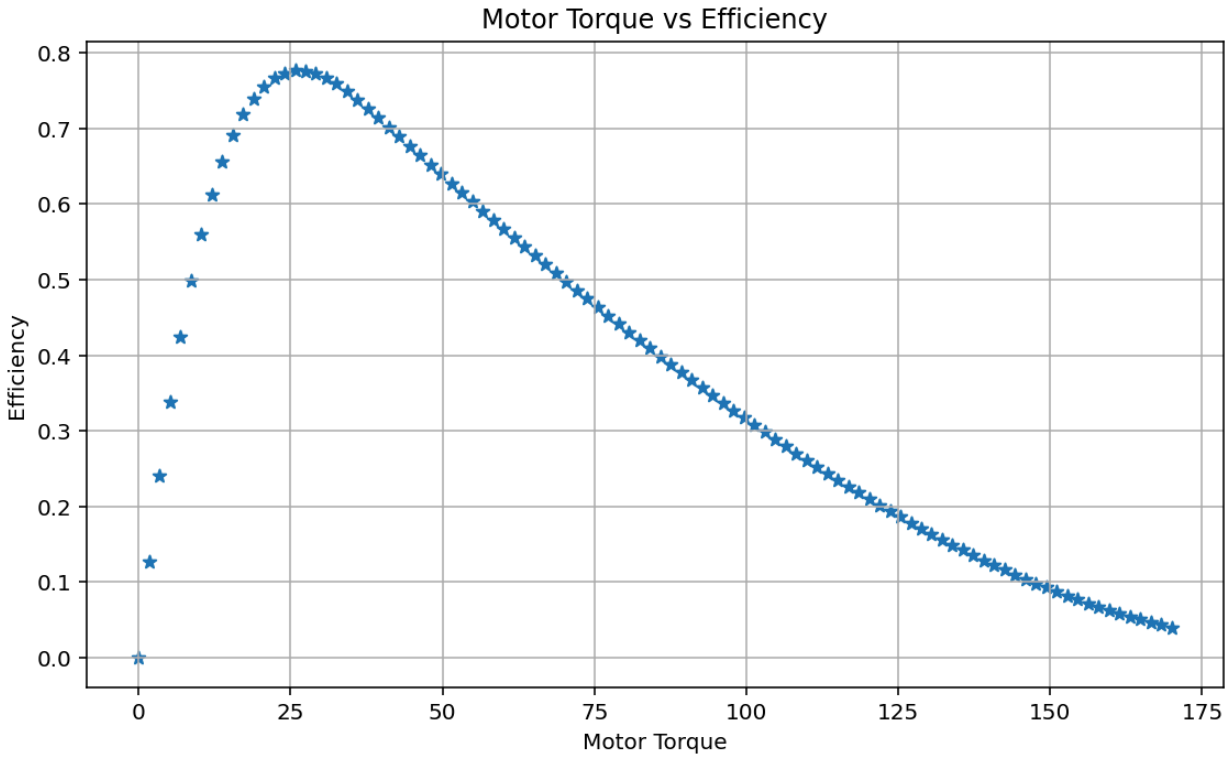


Figure 2: Motor Torque vs Efficiency Plot.

- Comparing Fig 2 and efficiency_experiment yields similar results that the rate of change is extremely steep from an initial jump in torque and maxes out early. Following maximum efficiency yields a slow decay to maximum torque of the motor. The team defined the efficiency function from our motor dictionary in subfunctions. The team verified the results from Fig 2 concluding that the efficiency of the motor is valid during experiment 1 and that it makes physical sense.

Task 8 Write Up:

a. The relationship between position and time appears to be linear, as shown by the straight line in Figure 3. However, a linear relationship between position and time means that velocity is constant. This contradicts the relationship we got in the second graph of Figure 3. One reason it likely appears linear is the small difference in velocity throughout the trip. The rover maintains approximately the same velocity throughout, as its velocity only ranges between 0.33 and 0.36 m/s. The graphing tool might also have some resolution problems making it hard for it to portray small differences like the ones we have obtained. Therefore, the graph of position vs. time is linear but in reality it is not.

b. The relationship between velocity and power seems to be inverse. This is interesting because power equals the product of torque and angular velocity, where angular velocity

is $(N_g V)/r$. However in Project 1, we learned that torque also decreases as angular velocity increases. This suggests that the decrease in torque is driving the inverse relationship between power and velocity. Therefore, as the speed increases the torque is decreasing thus causing the power to decrease as well.

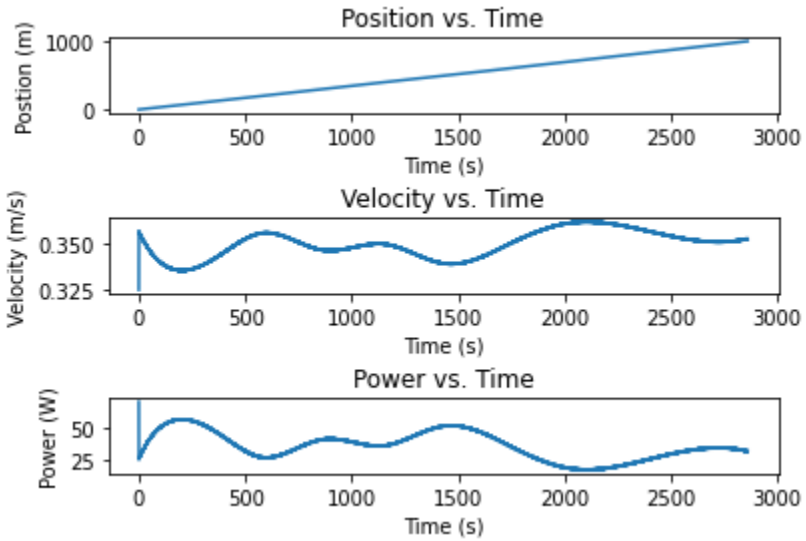


Figure 3: Position vs. Time, Velocity vs. Time, and Power vs. Time Graphs of Marvin

c. Using Figure 1, we can assume that the terrain angle has an inverse effect on the rover's velocity. This makes sense because as the terrain angle increases, the rover likely has a harder time traveling uphill. This is because the gravitational force acting directly against the rover increases. Thus, Marvin slows down. On the other hand, when the terrain angle is negative, the rover increases to its highest velocity because the force of gravity is acting in the direction of Marvin.

d. The velocity graph appears to be smooth. This makes sense because we used cubic interpolation to get the terrain angle which looks to make a smooth curve between points. Since, the forces that drive the velocity of Marvin are proportional to the smooth terrain angle we get a smooth velocity graph.

Table 1: Marvin Telemetry Data for experiment1

rover['telemetry']['...']	Value
completion_time	2859.48 seconds
distance_traveled	1000.0 meters
max_velocity	0.36235 m/s
average_velocity	0.34971 m/s
battery_energy	1128201.39 Joules = 1.128 x 10^{-6} Joules

batt_energy_per_distance	1128.20 J/m

Task 9 Write Up:

- The rover will be equipped with a .0972 x 10^{-6} Joules lithium battery. This amount of energy needed for the specific trip studied was 1.128 x 10^{-6} Joules as seen in Table 1. Since the value needed is greater than what is provided by the battery, the rover can not complete the case defined by experiment1. We obtained the value of battery_energy by integrating the result of $\frac{6\ x\ Power_{Mechanical}}{Efficiency}$ using trapezoidal rule. We multiplied the value by 6 to ensure we were getting the battery energy of the total system, not just each wheel.